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RESEARCH ARTICLE

NSNPSO-INC: A Simplified Particle Swarm Optimization Algorithm for Photovoltaic MPPT Combining Natural Selection and Conductivity Incremental Approach

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ABSTRACT To enhance photovoltaic (PV) system performance under various environmental conditions, this paper proposes a hybrid Maximum Power Point Tracking (MPPT) algorithm that integrates the Incremental Conductance (INC) method with an improved Particle Swarm Optimization (PSO) algorithm. The proposed approach simplifies the PSO velocity update process and uses natural selection to filter less adaptive particles. Leveraging the rapid convergence of the INC method, this hybrid algorithm effectively tracks the maximum power point (MPP). Simulation results show that under static partial shading conditions, the NSNPSO-INC algorithm achieves a stable MPPT efficiency of 99.9%, with a 68.18% faster convergence than conventional PSO and more stable performance than traditional INC. Under dynamic irradiance conditions, NSNPSO-INC adapts quickly, improving convergence speed by 93.33% over conventional PSO. Experimental results further demonstrate that NSNPSO-INC reaches steady state fastest, with an 85.52% improvement in convergence speed over traditional PSO. The algorithm maintains a stable power output around 550W even under cloudy conditions, significantly enhancing PV energy conversion efficiency and providing valuable insights for future MPPT algorithm development.

INDEX TERMS PV array, MPPT, PSO algorithm, boost converter.

I. INTRODUCTION

In recent years, the issues of environmental degradation and climate change have become significant challenges for the world, with increasing greenhouse gas emissions, rising global temperatures, and frequent occurrence of extreme weather, which significantly impact human society and natural ecology [1]. This set of environmental problems urgently requires people to find sustainable solutions to reduce dependence on fossil fuels and mitigate the rate of climate change. Among the many sustainable energy solutions, the development and utilization of clean energy have become a global consensus. In particular, PV power generation technology, as a way to directly convert solar energy into electricity, is considered one of the most important ways to address the current energy and environmental challenges due to its clean, renewable, and widely available characteristics [2], [3]. PV power generation technology converts solar energy into electricity using semiconductor materials, an efficient and environmentally friendly process. Nevertheless, the conversion efficiency of PV systems is affected by various factors, including weather conditions, temperature, and light intensity [4]. Especially in partially shaded conditions, it is difficult for PV power systems to obtain maximum output power through MPPT [5]. Therefore, in order to maximize the output power of a PV system, it is crucial to develop a technology that can adapt to environmental changes and continuously track the MPP.

The MPPT algorithm is an intelligent control algorithm designed to adjust the operating point of a PV system in

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real-time so that it can produce maximum output power under any given conditions. Implementing such algorithms is essential to improve the overall efficiency and economy of PV systems [6]. However, existing MPPT techniques face multiple challenges, including responsiveness to environmental changes, system complexity, and cost-effectiveness. Traditional MPPT methods such as perturbation observation (P&O) [7], [8] and incremental conductance (INC) algorithms [9] are widely used. They can track the MPP very efficiently with a reasonable convergence rate under normal conditions, i.e., uniform irradiation. However, these methods are unsuitable for the MPP due to the continuous oscillations that usually occur around the MPP due to frequent perturbation conditioning, and the oscillatory behavior leads to a steady state process with the power of considerable loss. At the same time, optimal performance may not be achieved under rapidly changing environmental conditions, especially in the case of drastic changes in light intensity and localized shadows due to fast-moving clouds. Therefore, developing a new MPPT algorithm that can quickly respond to environmental changes while maintaining system stability and high efficiency under localized shadows has become a hot topic of current research. With the wide application and rapid development of artificial intelligence (AI) technology in various fields, AI technology has become essential to solving the maximum power point problem in PV systems. Algorithms have been widely used in MPPT because of their excellent nonlinear mapping ability, self-learning, and adaptive capability. Reference [10], [11] used a neural network algorithm to track the MPP. In MPPT, the neural network can be trained to learn the performance of the PV system in different environments to predict the MPP. However, the disadvantage of neural network algorithms is that they require a large amount of historical data for training, which is time-consuming. Reference [12] used fuzzy logic control (FLC), which can adapt to the dynamically changing environment of PV to achieve faster tracking results. However, its drawbacks are that it requires expert knowledge to construct a fuzzy rule base, and its accuracy may be limited for complex or unknown system performance changes.

In addition, meta-heuristic intelligent algorithms that are naturally simulated in AI algorithms are a class of optimization algorithms inspired by nature that simulate mechanisms in natural processes such as biology, physics, or chemistry to solve complex optimization problems. They are widely used to solve various engineering problems, including MPPT, due to their simplicity, flexibility, and ease of implementation. Several bionic MPPT algorithms have been proposed in the: ant colony optimization (ACO) [13], particle swarm optimization (PSO) [14], cuckoo search (CSA) [15], flickering firefly [16], bat algorithm [17], and Gray Wolf Optimization (GWO) [18].

Among these algorithms, PSO is a very famous and effective meta-heuristic algorithm inspired by the observation of foraging behavior of bird flocks.The PSO algorithm is widely used in a variety of complex optimization problems due to its simplicity and ease of implementation, fast convergence, and strong global search capability. In the MPPT of PV system, the traditional algorithm is easy to fall into the local optimal solution, especially under the complex environmental conditions. Combining PSO and MPPT utilizes the global search capability of PSO, which can effectively avoid the limitation of traditional methods falling into local optimality, and can find the globally optimal maximum power point more accurately and quickly under multi-peak conditions. Therefore, the combination of PSO and MPPT has become an effective means to improve the output efficiency of PV systems and has been widely studied and applied.

In the field of PV MPPT, there have been numerous studies on the application of PSO variants [19] presents a multi-objective path planning for mobile robots based on an improved particle swarm algorithm to improve the global search capability, stability, and convergence speed. Reference [20] came up with a contraction factor PSO (CFPSO) algorithm that enhances the development phase of the SPSO algorithm by using a particular contraction factor (CF) in which each particle updates its position in tiny steps to avoid the particle moving outside the search space. However, the CF value is minimal. In that case, the particles cannot move far enough to obtain a better position within the search region since a smaller CF leads to relatively small particle velocities and, thus, longer tracking times for the GMPP. In the [21], the authors suggest using a linear time-varying function to tune the algorithm parameters, with inertial weight parameters and cognitive coefficients decreasing linearly with time, from high to low. On the contrary, the social coefficients increase linearly with time from low to high. However, linear time-varying values of inertia weight parameters and acceleration factors may not work effectively in localized shadows due to the distorted nonlinear nature of the PV array. Reference [22] describes a new strategy for determining PSO control parameters in order to minimize the convergence time and the failure rate of a PV maximum power point tracker system. The strategy determines these parameters offline, and then the control system uses them for the online MPPT. This strategy can be used not only for particle swarm optimization but also for all other optimization techniques.

With the development of AI technology, PSO is gradually combined with other intelligent algorithm optimization to further enhance its performance in PV MPPT. Reference [23] proposed a method combining PSO and genetic algorithm (GA) (PSO-GA) for optimizing fuzzy systems in MPPT controllers. The method optimizes the shape, affiliation function and fuzzy rules of the fuzzy method by PSO-GA to improve the speed and accuracy of the system. The experimental results show that the proposed method can reduce the steady state oscillations and improve the response speed and accuracy compared to other methods. Reference [24] A hybrid algorithm based on GWO and PSO is proposed, which improves the convergence speed and stability of PSO in complex environments by introducing the leadership mechanism of gray wolf optimization [25] introduced the grouping idea of shuffled frog leaping algorithm (SFLA) is introduced in the basic PSO algorithm (PSO–SFLA), ensuring the differences among particles and the searching of global extremum [26] analyzes the effectiveness of four intelligent methods in MPPT, including fuzzy logic (FL) and three meta-heuristic algorithms based on artificial neural networks, and evaluates the accuracy of the system using statistical metrics such as the root mean square error (RMSE) and the mean absolute error (MAE), which provide a comparison of the performance of the different intelligent algorithms in MPPT applications.

However, despite the fact that these PSO variants show significant improvements in PV MPPT, the traditional PSO and its variants still have some limitations when facing the strong nonlinear characteristics of PV systems and dynamic environmental changes. In particular, the PSO algorithm tends to fall into local optimal solutions under partial shadowing and rapidly changing weather conditions. Therefore, further research on how to improve the PSO algorithm to enhance its global search capability and adaptability in complex environments remains an important research direction.To address these issues, a simplified particle swarm optimized maximum power point tracking algorithm (NSNPSO-INC) combining natural selection and conductivity increment method is proposed in this paper. The algorithm enhances the global search capability of the algorithm by introducing the natural selection mechanism, which makes the particle swarm preferentially select the particles with higher adaptability for position updating in each generation iteration. Meanwhile, combining the advantages of the conductivity increment method improves the convergence speed and adaptability of the algorithm under dynamic light conditions, thus solving the limitations of the traditional PSO algorithm in photovoltaic systems.

Section II-A of this paper synthesizes previous studies, models the PV system under localized shading, and describes the fundamental principles of individual PV modules. Section II-B explores the effects of varying irradiance and temperature on the maximum power of the PV system under localized shading through simulation. Section 2.3 analyzes the traditional Incremental Conductance method, the conventional PSO algorithm, and the newly improved NSNPSO-INC algorithm proposed in this paper. It also introduces the selection criteria for algorithm parameters. In Section III-A, we describe several cases of boost converter parameter selection and simulation. Section III-B analyzes and discusses the simulation results, while Section III-C presents the experimental platform and results. Finally, Section IV summarizes the entire study and highlights the significance of the proposed algorithm.

II. MODEL ANALYSIS

A. MODELING THE SYSTEM IN LOCALIZED SHADOWS

In order to study the performance of the algorithm under particular shading and variable weather conditions, a power generation model of a PV system under partial shading is developed. The whole system is shown in FIGURE 1. In a natural PV power generation system, the whole PV array is usually composed of multiple solar panels connected in series and parallel, where each solar panel is connected in parallel with a reverse bypass diode to prevent reverse current from damaging the PV array and to ensure that the current flows from the PV array to the load in only one direction [27]. The output of the PV array is connected in parallel with the Boost circuit, and the boost output to the load or DC bus is accomplished through the boost circuit. The MPPT controller calculates the real-time power based on the sampled PV array output voltage V_{PV} and output current I_{PV}, and then accomplishes the tracking of the MPP through the improved PSO and outputs a PWM wave to control the switching tubes in order to control the power of the load.

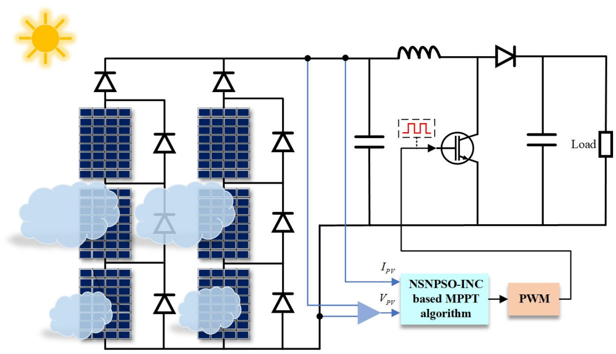

FIGURE 1. Mode of PV system under localized shading.

The performance of PV array output is closely related to weather variations, and it is critical to model PV cells properly to study the effect of weather conditions on PV output accurately. There is much available on modeling PV cells. Based on the diode five-parameter model established in the literature [28], the most widely modeled PV cell using a single diode model is shown in FIGURE 2.

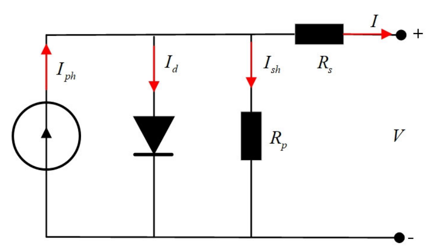

FIGURE 2. Single diode photovoltaic cell model.

The expression for the output current of the photovoltaic cell is [29] and [30]:

$$I_0 = I_{ph} - I_d(\exp[\frac{V_0 + IR_s}{vT}] - 1) - \frac{V_0 + IR_s}{R_p} \qquad (1)$$

where I_d is the diode saturation current, R_S is the series resistance, R_p is the parallel resistance, vT is the diode

TABLE 1. Parameters of photovoltaic solar modules.

Parameters	Values
Maximum power ($P_{\max}$)	213.15W
Open circuit voltage (V_{oc})	36.3V
Short-circuit current (I_{sc})	7.84A
Voltage at maximum power point (V_{mp})	29V
Current at maximum power point (I_{mp})	7.35A

ideal factor, I_0 and V_0 are the output current and voltage respectively, and I_{ph} is the photocurrent:

$$I_{ph} = I_{ph_{ref}}[1 + \alpha(T - T_{ref})]\frac{G}{G_{ref}} \tag{2}$$

where G is the irradiance, α is the relative temperature coefficient of the short-circuit current, T denotes the module temperature, and T_{ref} and G_{ref} denote the reference temperature and reference irradiance in the standard state (STC).

From Eq. (1) and Eq. (2), it is clear that the output of PV cells is closely related to both irradiance and temperature.

B. ANALYSIS OF PV OUTPUT UNDER LOCALIZED SHADING

Under unshaded conditions, the solar cells in a PV array exhibit uniformity, with bypass diodes remaining off as they bear reverse voltage, resulting in a single MPP for the array. However, when partial shading occurs, the output characteristics of the PV array become inconsistent. The short-circuit current of the shaded PV cells decreases, causing the bypass diodes to bear forward voltage and switch from off to on, thereby protecting the PV cells. Previous studies have demonstrated that the more complex the shading condition, the more varied the light intensity values received by the PV array, leading to an increased number of local maximum power points. This complexity can result in the failure of MPPT algorithms [31], [32].To investigate the impact of different irradiance and temperature conditions on the output of PV cells under partial shading, this paper establishes a simulation model of a PV array composed of three series-connected PV modules. The initial irradiance levels of the three PV modules are set to 1000 W/m^2, 600 W/m^2, and 400 W/m^2, respectively, with a temperature of 25° C. The parameters of the PV panels are listed in TABLE 1.

The output characteristic curves of the PV cells under various partial shading conditions are shown in FIGURE 3.

As illustrated in FIGURE 3(a) and FIGURE 3(b), the irradiance under localized shadowing increases gradually from 800 W/m^2 to 1400 W/m^2. With the increase in irradiance, the short-circuit current and the open-circuit voltage of the PV cell also increase, resulting in a higher overall output current and power. However, this leads to the emergence of multiple power peaks. In FIGURE 3(c) and FIGURE 3(d),

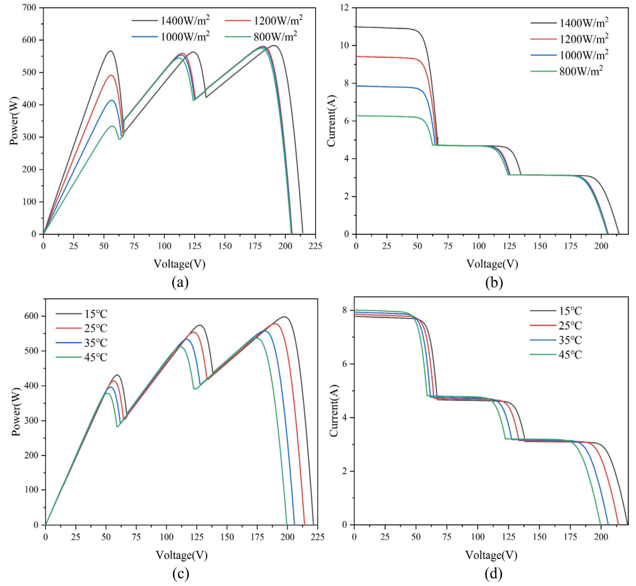

FIGURE 3. Photovoltaic PV and IV curves at different irradiance and temperature under localized shading (a) PV curves at different irradiance (b) IV curves at different irradiance (c) PV curves at different temperatures (d) IV curves at different temperatures.

it is evident that the temperature under localized shading rises from 15°C to 45°C. As the temperature increases, the short-circuit current of the PV cell decreases, while the open-circuit voltage increases, leading to a reduction in the overall output current and power. Despite these changes, multiple power peaks still occur. Therefore, variations in irradiance and temperature do not alter the localized impact on the PV cell's multiple power maxima. However, they do influence the overall maximum power point's rise and fall.

III. MPPT ALGORITHM

A. INC ALGORITHM

The INC algorithm [9] is a conventional method for realizing MPPT, and the core idea is to determine the location of the MPP by comparing the derivative of the ratio of current to voltage at the current operating point of the photovoltaic panel and its incremental change. The flowchart of the algorithm execution is shown in FIGURE 4.

The algorithm runs as follows:

Step 1: Set the initial voltage V_{0_INC} and initial current I_{0_INC}. This initial point is set to start the iterative process of the algorithm.

Step 2: In each iteration, the PV cell's current output voltage V_{INC} and current I_{INC} are measured. From this, the current power P_{INC} is calculated.

Step 3: Define the conductance G with respect to V_{INC} and I_{INC} as:

$$G = \frac{dI_{INC}}{dV_{INC}} \tag{3}$$

The amount of change in conductance is the difference between the current conductance and the last measured

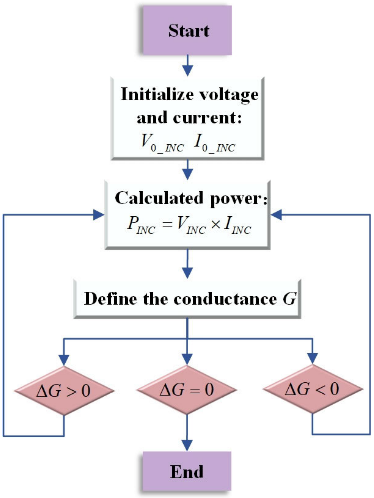

FIGURE 4. **Flowchart of INC algorithm.**

conductance, which is:

$$\Delta G = G - G' \tag{4}$$

where G' is the conductance value calculated in the previous iteration.

Step 4: When $\Delta G > 0$, it indicates that increasing the voltage will increase the power, and the operating voltage of the PV panel should be increased at this time.

When $\Delta G < 0$, it indicates that increasing the voltage will cause the power to decrease, and the operating voltage of the PV panel should be reduced at this time.

When $\Delta G = 0$, it indicates that the current is close to the MPP.

Step 5: Based on the above analysis process, the system adjusts the operating point by varying the operating voltage of the PV panel until the system stabilizes around the maximum power point.

Summarizing the above analysis, the successful implementation of INC relies on accurate voltage and current measurements and efficient control algorithms to adapt to changing environmental conditions. The advantages of INC are simple implementation and fast response time. However, the response may need to be faster in rapidly changing environments or oscillate in the case of small light changes. It affects the stability and efficiency of the system.

B. PSO ALGORITHM

PSO algorithm is a population intelligence algorithm inspired by the foraging behavior of bird flocks. PSO algorithm was first proposed by Kennedy and Eberhart in 1995 [33], and the core idea of PSO is to solve the optimization problem by simulating the social behaviors of bird flocks during foraging. In this model, each bird is abstracted as a "particle" in the search space, and each individual in the swarm can memorize the optimal position it has found and, at the same time, know the optimal position found by the whole group. By sharing information between individuals, the whole flock can collaborate to find the global optimal solution in the multidimensional search space. The iterative formula of the PSO algorithm is [34]:

$$\begin{cases} v_i(k+1) = w \cdot v_i(k) + m_1 r_1 [P_i - x_i(k)] \\ \qquad + m_2 r_2 [P_u - x_i(k)] \\ x_i(k+1) = x_i(k) + v_i(k+1) \end{cases} \tag{5}$$

where w is the weight coefficient reflecting the size of the particle's inertia; m_1 and m_2 are the learning factors reflecting the particle's self-cognitive ability and social cognitive ability, respectively, and controlling the tendency of the particle to move to the individual optimal and global optimal position; r_1 and r_2 are random numbers between 0 and 1 to increase the particle's randomness in the search space; there are n particles in the space, and each particle corresponds to a position x_i and a velocity value v_i, and all the particles in the space combine and compete with each other to find the optimal position. A velocity value v_i, all the particles in the space combine and compete with each other to search for the optimal position, and the optimal position P_i of each particle and the global optimal position P_u are updated according to the fitness of the particles. The update of each particle's velocity magnitude v_i (k+1) depends on the current velocity v_i (k), individual optimal position P_i, and global optimal position P_u.

When applying PSO to MPPT, the switch duty cycle represents the particles' positions, while the output power at each position indicates their fitness. The algorithm iteratively updates particle positions, enabling the population to converge on the optimal solution and effectively track the MPP. PSO's ability to share information among particles enhances its precision and avoids local optima. However, this requires careful parameter tuning. Under partial shading, the PV array's multiple peaks can trap PSO in local optima. Rapid weather changes can also shift the MPP, and the conventional PSO's velocity update rules may limit convergence speed, making it difficult to adapt to these dynamics promptly.

C. NSNPSO ALGORITHM

In the basic PSO algorithm,the random numbers r1 and r2 are introduced primarily to enhance the partices'ability to explore a broader search space during the initial stages of optimization,thereby increasing the likelihood of identifying a globally optimal solution.However,when applying the PSO algorithm to MPPT in PV systems,the persistence of r1 and r2 values close to 1 in the later stages of the algorithm may result in excessive particle velocities,which could cause the particles to overshoot the maximum power point.To improve accuracy and stability in the MPPT process,it is advisable to gradually reduce or even eliminate the influence of r1 and r2 as the optimization progresses,thereby mitigating the risk of excessive velocity. By directly leveraging the power-voltage(P-V)characteristics for MPPT control and removing the impact of random numbers r1 and r2.the tracking process can be made more stable,minimizing the potential for overshooting the maximum power point due to high speedsin the later stages.

The iterative equation of the simplified PSO algorithm after removing the stochastic parameters is:

$$\begin{cases} \Delta d_i(k+1) = w \cdot \Delta d_i(k) + m_1[d_{ibest} - d_i\,(k)] \\ \quad +m_2[d_{gbest} - d_i\,(k)] \\ d_i(k+1) = d_i(k) + \Delta d_i(k+1) \\ d_{ibest} = d_i(k), P_i(k) > P_{i\max} \\ d_{gbest} = d_i(k), P_i(k) > P_{g\max} \end{cases} \tag{6}$$

where P_i(k), d_i(k), Δd_i(k) are the power, duty cycle, and the amount of change in the duty cycle of the ith particle, respectively, the position of the particle is defined as the duty cycle d_i, the particle velocity represents the change amount of the duty cycle Δd_i. The output power P_i corresponding to each duty cycle is used as the adaptation degree of the particle. Update the particle's optimal duty cycle digest and global maximum output power P_{gmax} according to the particle's current maximum power P_{imax}, and then update the global optimal duty cycle best d_{gbest}. Each particle adjusts the change amount of the duty cycle in the next iteration according to the inertia, the individual optimal duty cycle digest, and the global optimal duty cycle best d_{gbest}, which updates the particle's duty cycle in real-time d_i (k+1).

To address the common issues of search efficiency degradation and premature convergence in the later stages of PSO algorithms, current improvement strategies often involve introducing new computational procedures in the algorithm's later stages to enhance optimization capabilities and effectively avoid entrapment in local optima. In contrast, this dissertation proposes an innovative method focused on the early-stage particle selection process, drawing inspiration from the natural selection theory observed in biological systems. This concept is integrated into the MPPT method within traditional PSO algorithms. Specifically, prior to updating particle velocities, the proposed method selects the top 50% of particles with the highest fitness in each iteration, while eliminating the bottom 50% with lower fitness. The advantage of this approach lies in its selective retention of higher fitness particles, thereby accelerating both particle updates and the generational evolution of the population. Although this selection strategy reduces the number of particles per iteration, the population's diversity and size are preserved through appropriate complementary mechanisms, such as the introduction of new particles or the replication of highly adapted particles. Consequently, this approach not only maintains the memory of particles' historical optimal positions and simplifies the velocity update process but also effectively mitigates the risk of premature convergence, thereby enhancing the accuracy and efficiency of the optimization process.

D. PARAMETER DESIGN

Whether the PSO algorithm can accurately track the maximum power point, especially in the case of local shadowing, the values of inertia coefficients w, self-cognition coefficients m_1, and social cognition coefficients m_2 have a powerful influence on the tracking process of the maximum power point [35]. Therefore, based on the existing research, this paper uses the adaptive adjustment method of PSO parameters.

The larger the inertia weight coefficient w, the larger the particle's initial velocity in space, and the worse the convergence accuracy is. In the initial stage of the MPPT tracking process, w should take a more significant value, which is conducive to the fast search in the global range to avoid missing the information of the possible maximum power points; as the number of iterations increases, the particle has already been aggregated in the range of the global maximum power points in the later stage of iteration. Then, the value of w should be smaller so that the particles move slower to stabilize near the global maximum power. Therefore, in this paper, we refer to the sine-cosine algorithm [36] and use the cosine decreasing inertia weights:

$$w = w_{\min} + (w_{\max} - w_{\min})\cos(\frac{k}{N}) \tag{7}$$

where k is the current iteration number, N is the maximum value of the iteration number, and $\mathrm{w}_{\max}$ and $\mathrm{w}_{\min}$ are the maximum and minimum inertia weight coefficients, respectively.

From the [31], the larger the self-cognition coefficient m_1 is, the stronger the self-cognition ability of the particles, and the stronger the ability to track the local maximum power points; the more significant the social cognition coefficient m_2 is, the stronger the overall social cognition ability of the particles, and the stronger the ability to converge to the global maximum power points. Therefore, in the initial stage of the MPPT tracking process, m_1 should take a more significant value, and m_2 should take a smaller value so that all particles can find all the local maximum power points as much as possible, avoiding omission and at the same time, not misjudging the local maximum power point as the global maximum power point and converging prematurely. At the late stage of tracking, m_1 should take a smaller value, and m_2 should take a more significant value so that the particles gather towards the global maximum power point and the convergence is completed quickly. Based on this, [37] proposed the method of time-varying acceleration coefficient:

$$\begin{cases} m_1 = (m_{1f} - m_{1i})\dfrac{k}{N} + m_{1i} \\ m_2 = (m_{2f} - m_{2i})\dfrac{k}{N} + m_{2i} \end{cases} \tag{8}$$

where m_{1i}, m_{1f}, m_{2i}, m_{2f} are fixed constants, k is the current number of iterations; N is the maximum value of the number of iterations.

In addition, the selection of the initial population also has an essential impact on whether the whole algorithm can find the optimal smoothly and quickly, and the random initialization of the population may lead to a longer convergence time and a higher failure rate [38]. To avoid this problem, the initial particle population can be uniformly distributed in the range of [0.1, 0.9], and the initial individual assignment of the particle population can be carried out by Eq. (9) [39]:

$$d_0^k = \frac{k}{(ss+1)} \tag{9}$$

where d_0^k denotes the value of the initial particle (duty cycle), ss is the population size, and k denotes the serial number of the particle in the population ($k = 1,2,3\ldots ss$).

In order to ensure the fairness and authenticity of the simulation and experimental comparisons, the results of each algorithm are run under the same parameter criteria. The settings of the specific parameters are:$n= 8$,$w_{min} = 0.2$,$w_{max} = 0.9$,$m_1 = 1.5$,$m_2 = 1.5$, n is the initial number of particles.

E. COMPOSITE MPPT ALGORITHM (NSNPSO-INC)

The INC algorithm is widely recognized for its quick convergence and straightforward implementation, making it highly efficient in stable or moderately fluctuating PV environments. However, when faced with highly dynamic and rapidly changing conditions, such as sudden variations in solar irradiance or temperature, the INC algorithm can struggle to maintain optimal tracking. It may quickly converge to a local maximum, especially in complex scenarios with multiple peaks, leading to suboptimal performance.

On the other hand, the NSNPSO algorithm excels in its ability to explore the global search space effectively, thereby reducing the likelihood of being trapped in local optima. This makes NSNPSO particularly valuable in scenarios where the PV system's characteristics are complex and exhibit multiple local maxima. Despite its robustness in finding the global optimum, the NSNPSO algorithm may exhibit slower convergence compared to the INC algorithm, especially when the system's conditions are stable or change gradually. In the context of a highly dynamic PV environment, where both rapid response and robust global optimization are crucial, relying solely on either the INC or NSNPSO algorithm presents significant limitations. The INC algorithm's speed is offset by its vulnerability to local optima, while NSNPSO's global accuracy is hindered by slower convergence.

To address these challenges, this paper proposes a composite MPPT algorithm that integrates the quick convergence of the INC algorithm with the global search capability of the NSNPSO algorithm. By combining these two approaches, the proposed algorithm leverages the strengths of both: the INC component ensures rapid tracking in response to sudden environmental changes, while the NSNPSO component continuously fine-tunes the search to avoid local maxima and achieve global optimization. This synergistic approach allows the algorithm to effectively manage the trade-offs between convergence speed and optimization accuracy, ensuring more reliable and efficient MPPT performance even in the most challenging PV system conditions. The operation process of this algorithm is illustrated in FIGURE 5.

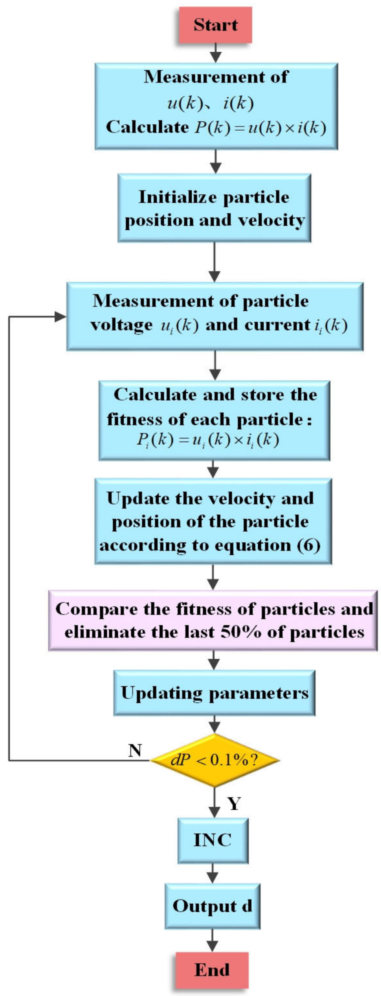

FIGURE 5. NSNPSO-INC control flowchart.

The algorithm runs as follows:

Step1: Detect the current voltage $u(k)$ and current $i(k)$ of the PV arrays and calculate the power out of the output power according to Eq. (10):

$$P(k) = u(k) \times i(k) \tag{10}$$

Step 2: Initialize the position and velocity of the particle according to Eq. (9).

Step 3: Measure the voltages $u_i(k)$ and $i_i(k)$ of individual particles in the current population.

Step4: The current adaptation (output power) of each particle is calculated according to Eq. (11):

$$P_i(k) = u_i(k) \times i_i(k) \tag{11}$$

Step 5: Update the velocity and position of the particles. According to Eq. (6), the individual optimal duty cycle d_{ibest}

and the global optimal duty cycle d_{gbest} are updated, and the duty cycle variation $\Delta d_i(k+1)$ and $d_i(k+1)$ are updated for each particle.

Step 6: The fitness of each particle is compared with the historical optimum, and the particles of the current population are sorted according to the size of the fitness, eliminating the particles with the last 50% of fitness.

Step 7: Updating parameters. Update the inertia weight coefficient w, the self-perception coefficient m_1 and the social perception coefficient m_2 according to Eq. (7) and Eq. (8).

Step 8: Determine whether the termination condition is reached. This is done by calculating the transformation rate of two neighboring powers according to Eq. (12):

$$dP = \frac{|P_i(k) - P_i(k-1)|}{P_{g\max}} \tag{12}$$

When $dP < 0.1\%$ is satisfied, the NSNPSO algorithm is considered close to convergence at this point, and vice versa, go back to step 3 and iterate again.

Step 9: The conductance increment method is used to quickly complete the final convergence trace and output the duty cycle d.

The pseudo-code of the NSNPSO-INC algorithm is shown in Table 2:

TABLE 2. NSNPSO-INC algorithm pseudo code.

```
Step 1:
    Detect current voltage u(k) and current
i(k) of PV arrays
    Calculate output power P(k) using Eq. (10)
Step 2:
    Initialize particle positions and
velocities using Eq. (9)
Step 3:
    For each particle i in the population:
        Measure particle voltages ui(k) and
currents ii(k)
Step 4:
    For each particle i:
        Calculate current adaptation (output
power) Pi(k) using Eq. (11)
Step 5:
    For each particle i:
        Update velocity vi(k+1) using Eq. (6)
        Update position di(k+1) using updated
velocity
        Update individual best dibest and
global best dgbest duty cycles
Step 6:
    Compare each particle's fitness with its
historical best:
        If current fitness > historical best:
            Update historical best with current
fitness
    Sort particles by fitness
    Eliminate bottom 50% of particles with
lowest fitness
    If population size < initial size:
        Introduce new particles or replicate
top performing particles to maintain population
size
Step 7:
    Update parameters:
        Update inertia weight w using Eq. (7)
        Update self-perception coefficient m1
and social perception coefficient m2 using Eq.
(8)
Step 8:
    Check termination condition:
        Calculate power change rate ΔP between
two neighboring iterations using Eq. (12)
        If ΔP < 0.1%, convergence is assumed;
otherwise, return to Step 3
Step 9:
    Apply the conductance increment method to
refine convergence
    Output the optimal duty cycle d.
```

F. ALGORITHM COMPLEXITY ANALYSIS

Time complexity measures the time required for the execution of the algorithm and is usually expressed as a function of the input size. For PSO and NSNPSO-INC, it can be analyzed in the following steps:

PSO Time Complexity:

Step1: Initialization: Initialize particle positions and velocities with time complexity O(n), where n is the number of particles.

Step2: adaptation calculation: in each iteration, calculate the adaptation of each particle with a time complexity of O(n).

Step3: update particle position and velocity: for each particle, update its velocity and position, with a time complexity of O(n).

Step4: number of iterations: the maximum number of iterations is N. The total time complexity is O(N n).

NSNPSO-INC Time Complexity:

Step1: initialization: similar to PSO with O(n) time complexity.

Step2: adaptation calculation: in each iteration, the time complexity is still O(n).

Step3: natural selection mechanism: particles need to be selected, copied or eliminated, the complexity of this step is O(n log n) with respect to the number of particles (using sorting for selection).

Step4: conductivity increment method: this may involve additional calculations for each particle, assuming a time complexity of O(n).

Step5: updating particle positions and velocities: similar to PSO, with a time complexity of O(n).

Step6: total time complexity: due to the introduction of natural selection and the conductivity increment method, the time complexity of NSNPSO-INC is O(T n log n).

Spatial Complexity Analysis:

PSO spatial complexity: mainly store the position, velocity and fitness of each particle, the spatial complexity is O(n). NSNPSO-INC Spatial complexity: In addition to storing the position, velocity and fitness of each particle, the

TABLE 3. Boost converter rating parameters.

Component	Value
Capacitor (C_{in})	500uF
Capacitor (C_{out})	2000.255uF
Inductor(L)	8.578mH
Switching frequency (f)	20kHz
Load resistance (R)	20Ω
Sampling Time	2s

information related to the natural selection mechanism and the conductivity increment method is also stored, but the overall spatial complexity is still O(n). This is because despite the introduction of the natural selection and INC mechanisms, the additional storage requirements for these mechanisms are only linearly related to the number of particles and do not lead to an increase in space complexity. From the above analysis, it can be seen that NSNPSO-INC has a slightly higher time complexity than conventional PSO, mainly due to the introduction of the natural selection mechanism, which increases the complexity to O(n log n) instead of O(N n). However, the performance improvement brought by this complexity increase may be worthwhile, which can be supported by the following experimental results.

IV. SIMULATION AND EXPERIMENTAL VERIFICATION

A. SIMULATION MODEL ANALYSIS

In order to verify the feasibility of the proposed algorithm, the corresponding PV system is modeled in this paper using Matlab/Simulink. The energy conversion part uses a boost converter to convert the electrical energy generated by the PV array.

The specific parameters of the converter are shown in TABLE 3.

The overall PV array uses a 3 × 3 parallel combination, and the parameters of the PV modules are shown in Table 1. In order to test the tracking effect of the algorithm in different weather and shading situations, the temperature is always set to 25°C. Three different irradiance scenarios are shown in FIGURE 6, where the irradiance is in W/m^2.

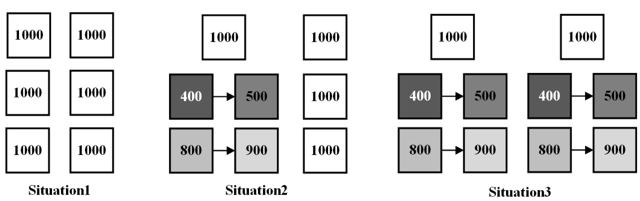

FIGURE 6. Three different irradiance scenarios.

In order to verify whether the algorithm can accurately track the maximum output power, the maximum output power in three different irradiance cases was simulated and calculated before starting the test. Their PV curves are shown in FIGURE 7, from which it can be seen that the maximum power in the three cases is 2553.3W, 1964.2W, and 1580.5W, respectively.

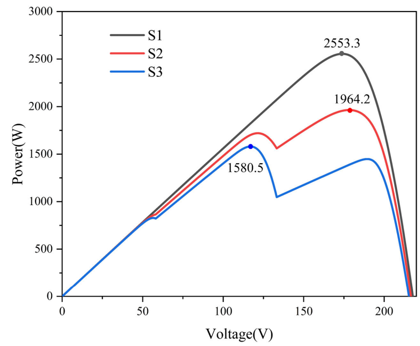

FIGURE 7. PV curves for the three scenarios.

This paper evaluates algorithms using two primary metrics: convergence time and tracking efficiency. The convergence time can be accurately obtained from the power waveform comparison graph when different algorithms reach the steady state. The tracking efficiency can be calculated by Eq. (13):

$$\eta = \frac{P_{mpp}}{P_{\max}} \times 100\% \tag{13}$$

where P_{mpp} is the output power when steady state is reached and P_{max} is the maximum output power of the PV array.

B. SIMULATION RESULTS AND DISCUSSION

Based on the three cases described in the previous section, this section verifies the power and duty cycle tracking effects in these cases using INC, PSO, PSO-INC, and NSNPSO-INC, respectively.

Situation 1: FIGURE 8 compares the tracking effectiveness of the four algorithms under static irradiance with no shadow influence, while FIGURE 9 presents statistical graphs of each algorithm's convergence time and tracking efficiency in this scenario. All algorithms successfully track the maximum power point, with the final converged power close to 2553.3 W. The INC algorithm converges the fastest at 0.06 seconds, followed by NSNPSO-INC at 0.14 seconds, PSO-INC at 0.29 seconds, and the conventional PSO algorithm at 0.44 seconds. All four algorithms stabilize the duty cycle around 0.2. However, the INC algorithm exhibits significant duty cycle fluctuation after reaching a steady state, unlike the more stable behavior of the other three algorithms. Although the INC algorithm excels in convergence speed, it continues to adjust and fluctuate in a steady state, leading to higher power loss. In contrast, the composite algorithms, particularly PSO-INC and NSNPSO-INC, demonstrate better stability and efficiency. NSNPSO-INC converges 68.18% faster than the conventional PSO algorithm. As shown in Fig.9, the MPPT tracking efficiency of PSO-INC and NSNPSO-INC is around 99.99%, whereas the high fluctuation of the INC algorithm results in relatively lower efficiency due to increased power loss.

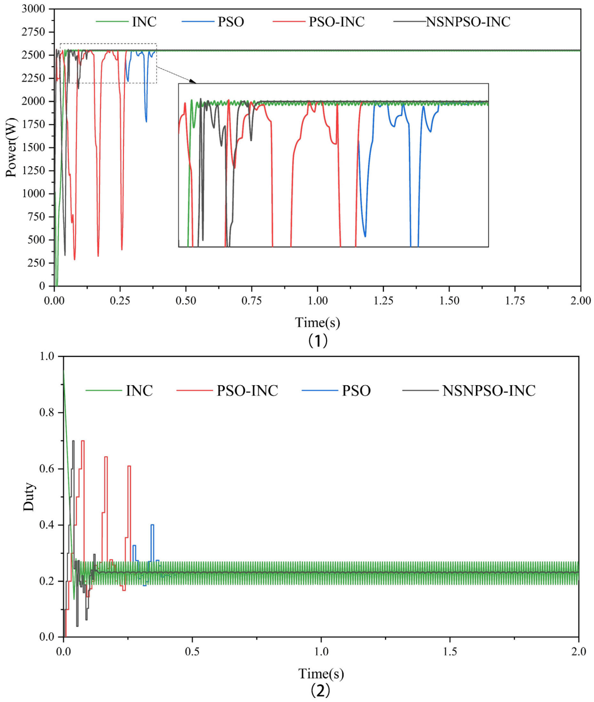

FIGURE 8. **Tracking performance for all methods in S1(1)Comparison of power waveforms at S1(2)Duty cycle waveform comparison at S1.**

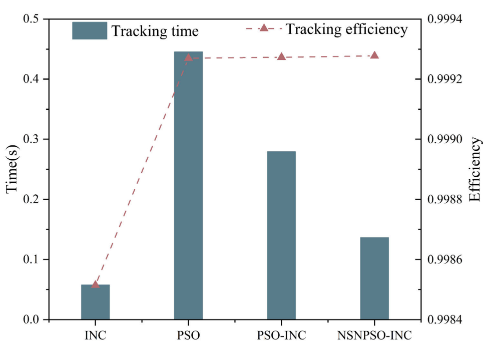

FIGURE 9. **Tracking time and tracking efficiency at S1.**

Situation 2: The results illustrated in FIGURE 10 and FIGURE 11 clearly demonstrate the superior performance of the NSNPSO-INC algorithm under dynamic conditions, particularly in the presence of localized shadows on the PV array. Initially, the INC algorithm exhibits the fastest convergence speed between 0 and 1 second, aligning with the behavior observed in Case 1. However, despite its quick response, the INC algorithm fails to accurately track the MPP, stabilizing at a suboptimal power output of 1659.8 W compared to the true maximum of 1826.1 W. This discrepancy highlights a significant limitation of the INC algorithm: its rapid convergence comes at the cost of precision, particularly in complex environments where local maxima can mislead the algorithm.

In contrast, the NSNPSO-INC algorithm, while slightly slower than the INC algorithm in the initial phase, achieves a far superior balance between speed and accuracy. It not only converges faster than the other PSO-based algorithms but also achieves a power output much closer to the true maximum, indicating its enhanced precision in tracking the MPP. The algorithm's ability to stabilize quickly and accurately near the maximum power point is particularly evident after the abrupt irradiance change at 1 second, where the NSNPSO-INC algorithm re-stabilizes in just 0.03 seconds—93.33% faster than the conventional PSO algorithm.

Moreover, the NSNPSO-INC algorithm's performance during the 1-2 second period further underscores its advantages. Despite the sudden increase in irradiance, which shifts the MPP to 1965.2 W, the NSNPSO-INC algorithm efficiently tracks and stabilizes near this new MPP with remarkable speed and accuracy. In contrast, the INC algorithm, though responsive, fails to adjust effectively, stabilizing at a much lower power output of 1716.5 W. This behavior demonstrates the INC algorithm's susceptibility to rapid changes in environmental conditions, where its initial speed is offset by its inability to maintain high tracking efficiency under dynamic scenarios.

The statistical analysis presented in FIGURE 11 reinforces these observations. The NSNPSO-INC algorithm consistently outperforms the other algorithms in terms of both convergence time and tracking efficiency. Its ability to maintain an efficiency close to 99.99% across both the 0-1s and 1-2s periods highlights its robustness in handling complex, rapidly changing PV conditions. This superior performance is attributed to the algorithm's synergistic combination of the fast convergence characteristic of the INC algorithm with the global search robustness of the NSNPSO algorithm. This ensures not only quicker adaptation to changing conditions but also a higher degree of accuracy in maintaining optimal power output.

In summary, the NSNPSO-INC algorithm demonstrates a clear advantage in scenarios involving dynamic changes and environmental perturbations. Its ability to quickly converge while avoiding local optima, combined with its enhanced tracking precision, makes it a highly effective solution for maximizing power output in complex PV systems. This performance improvement is particularly critical in real-world applications, where the ability to swiftly and accurately respond to environmental changes directly impacts the overall efficiency and stability of the photovoltaic system.

Situation 3: FIGURE 12 compares the tracking effectiveness of the four algorithms under dynamic changes with localized shadows on all PV arrays, and FIGURE 13 presents the statistics of convergence time and tracking efficiency for this scenario. Initially, the irradiance for two columns of PV modules is set to 1000 W/m^2, and for the second and third modules, 400 W/m^2 and 800 W/m^2, respectively. At 1 second,

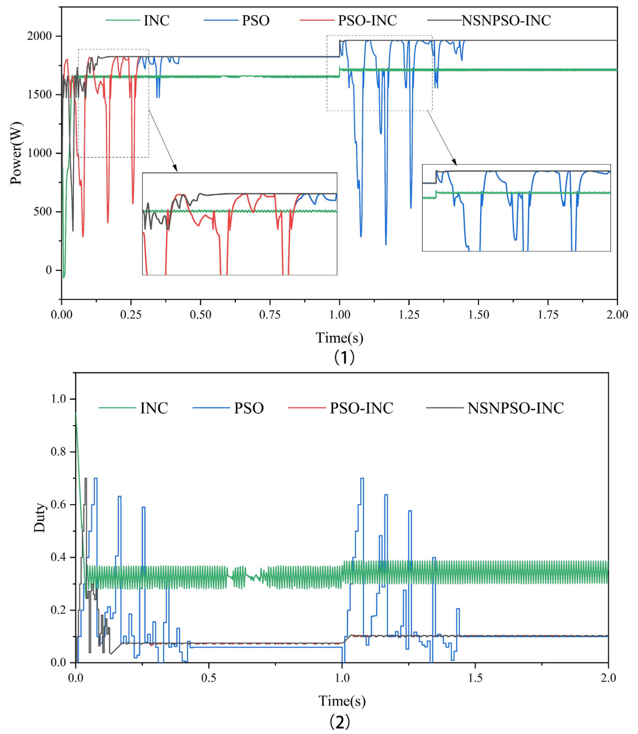

FIGURE 10. **Tracking performance for all methods in S1 (1) Comparison of power waveforms at S2 (2) Duty cycle waveform comparison at S2.**

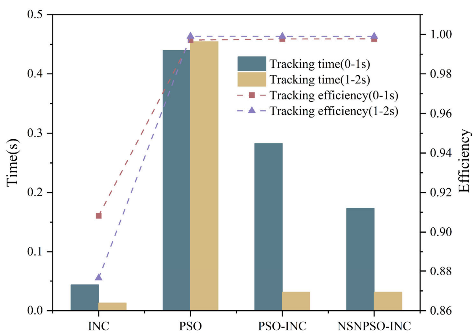

FIGURE 11. **Tracking time and tracking efficiency at S2.**

the irradiance for these four modules jumps to 500 W/m^2 and 900 W/m^2.

During the static period (0-1s), the convergence speeds of the four algorithms are consistent with the previous cases, but the INC algorithm suffers from a more severe local optimum issue. NSNPSO-INC demonstrates the best convergence speed while maintaining tracking accuracy and shows less fluctuation in duty cycle compared to the conventional PSO and PSO-INC algorithms.

After the sudden irradiance change at 1 second, NSNPSO-INC and PSO-INC quickly adjust and rediscover the maximum power point near 1580.5 W. The INC algorithm also finds the maximum power point again due to the irradiance change. During the 0-1s period, the INC algorithm's tracking efficiency is poor due to falling into a local optimum, while the other three PSO algorithms show similar tracking efficiencies. During the dynamic change from 1-2s, the tracking efficiency of all four algorithms stabilizes at 99.99%. The convergence speed and tracking efficiency in this scenario are similar to those in case 2, with the notable difference that the INC algorithm eventually returns to the maximum power point after the irradiance change.

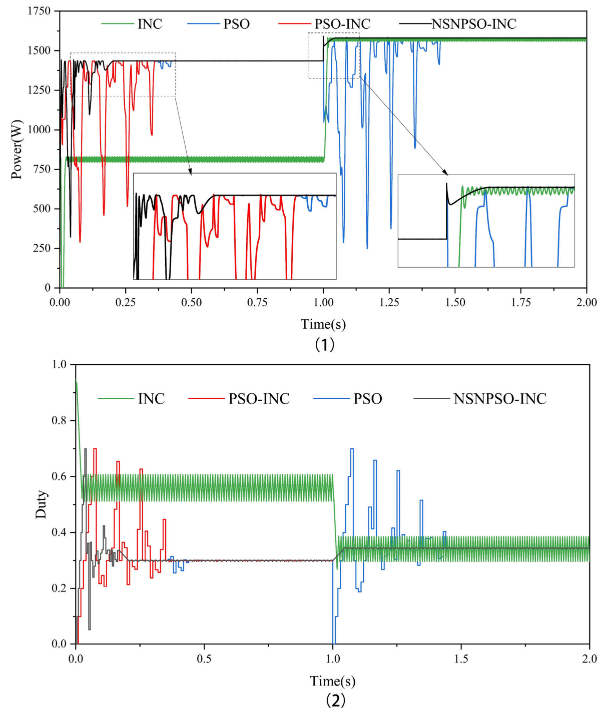

FIGURE 12. **Tracking performance for all methods in S3(1)Comparison of power waveforms at S3(2)Duty cycle waveform comparison at S3.**

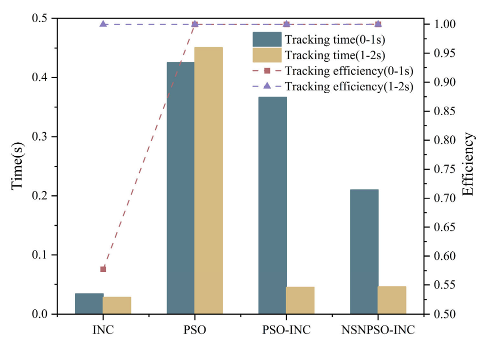

FIGURE 13. **Tracking time and tracking efficiency at S3.**

Combining the simulation results of the above three cases, the conventional INC algorithm performs the best in terms of convergence speed, but it may fall into the local optimal solution when facing the local shadows; in contrast, the PSO algorithm can avoid the situation of local shadows, but the convergence speed of the conventional PSO algorithm is too slow; the use of the PSO-INC and NSNPSO-INC algorithms has a significant improvement in terms of both

TABLE 4. Comparison table of different PSO algorithms.

Documentary	Circuit Type	algorithm	MPPT efficiency	track time
This paper	Boost	NSNPSO-INC	99.9%	0.14s
[23]	Boost	PSO-GA	99.9%	0.2s
[24]	Boost	GWO-PSO	99.9%	0.22s
[27]	Boost	OD-PSO	99.8%	0.18s
[40]	Boost	TSA-PSO	99.8%	0.4s
[41]	Boost	ELPSO-P&O	99.9%	0.2s
[42]	Boost	NF-PSO	99.6%	0.2s

convergence speed and tracking accuracy. Speed and tracking accuracy, especially the simplified particle swarm NSNPSO-INC algorithm combined with natural selection, has faster convergence speed, higher tracking efficiency, and better robustness under static and dynamically changing irradiance. In order to compare the performance of the proposed algorithms more comprehensively, this paper synthesizes the existing studies and conducts a comparative analysis, as shown in Table 4.

From the table, it can be seen that the NSNPSO-INC algorithm has superior convergence speed and tracking efficiency compared to several other PSO algorithms.

C. EXPERIMENTAL RESULTS AND DISCUSSION

In order to further verify the effectiveness of the proposed algorithm, this paper builds a set of PV boost converter experimental platforms based on the existing experimental conditions as shown in FIGURE 14.

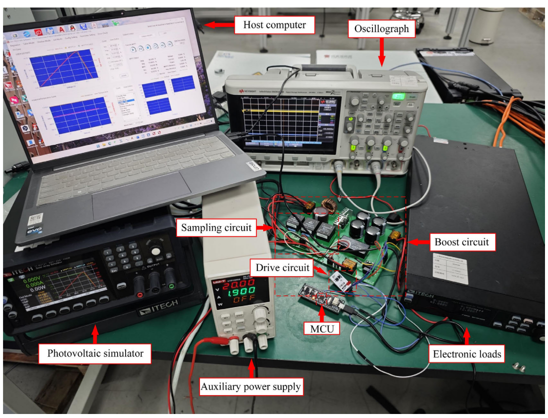

FIGURE 14. Photovoltaic DC-DC experimental platform.

The inductance of the boost converter is 220uH, the output filter capacitance is 2000uF, and the frequency is 20 kHZ

TABLE 5. Photovoltaic array parameters.

Maximum power	550W
Open circuit voltage	49.59V
Short-circuit current	13.82A
Voltage at MPP	41.55V
Current at MPP	13.24A

[43]. The input side uses a PV simulator to simulate the parameters of a commonly used PV array, as shown in TABLE 5.

On the output side, an electronic load is used to regulate the load voltage to a constant voltage mode so that the output side maintains a load voltage of 48V. The whole operation process is as follows: firstly, the input voltage and current are simulated by the PV simulator, and then the external power supply is used to supply power for the sampling short circuit; the sampling circuit sends the collected input voltage and current to the main controller after conversion, the main controller compiles and runs the MPPT algorithm and outputs the control signals to the driver circuit, and finally the PWM wave output from the driver circuit controls the switching of the switching tubes in the boosting circuit, which completes the tracking of the maximum power point and the tracking of the maximum power point, and the control signal is output to the driver circuit. Finally, the PWM wave output from the driver circuit controls the switching of the switching tube in the boost circuit to complete the tracking of the maximum power point and output a stable voltage.

In order to compare the effect of the algorithms proposed in this paper more intuitively, the experiment burned three MPPT algorithm programs of PSO, PSO-INC, and NSNPSO-INC in the central controller to compare their respective output waveforms. The waveforms of output voltage and output current were recorded using an oscilloscope. The convergence time and steady-state values were captured, and the experimental results are shown in FIGURE 15 From the figure, it can be seen that the convergence times of the three algorithms are 366ms (PSO), 254ms (PSO-INC) and 52.8ms (NSNPSO-INC) respectively, and the NSNPSO-INC algorithm has the fas'test convergence speed, PSO-INC is the second fastest, and PSO is the slowest. Based on the captured voltage-current steady-state data, the power of the three algorithms after reaching steady state is calculated to be 549.44 W (PSO), 545.96 W (PSO-INC) and 548.47 W (NSNPSO-INC), respectively, and the output power is near the maximum power. Therefore, the experimental results are consistent with the results obtained from the simulation, which further proves the reliability of the NSNPSO-INC algorithm.

The experiment simulates cloudy weather conditions using a PV simulator. FIGURE 16(a) illustrates the changes in

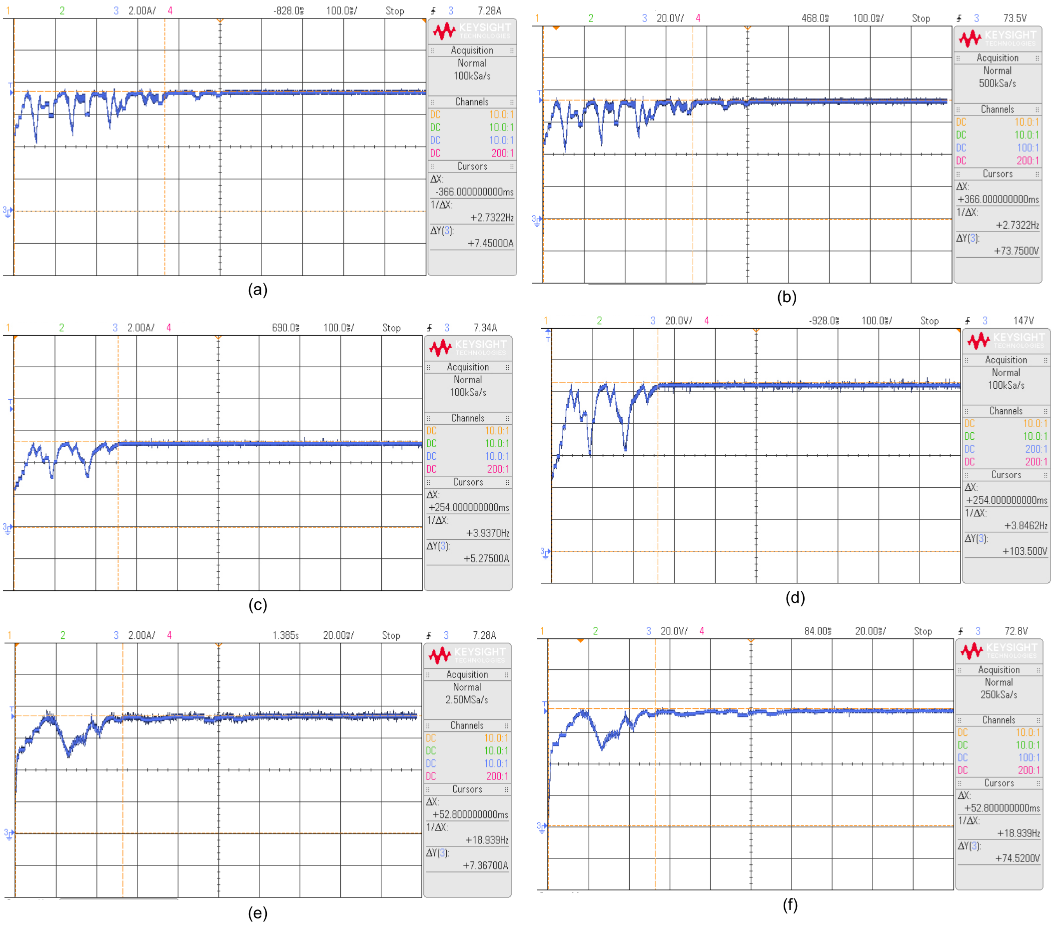

FIGURE 15. **Experimental waveforms: (a) Output current waveform using PSO (b) Output voltage waveform using PSO(c) Output current waveform using PSO-INC (d) Output current waveform using PSO-INC(e) Output current waveform using NSNPSO-INC (f) Output current waveform using NSNPSO-INC.**

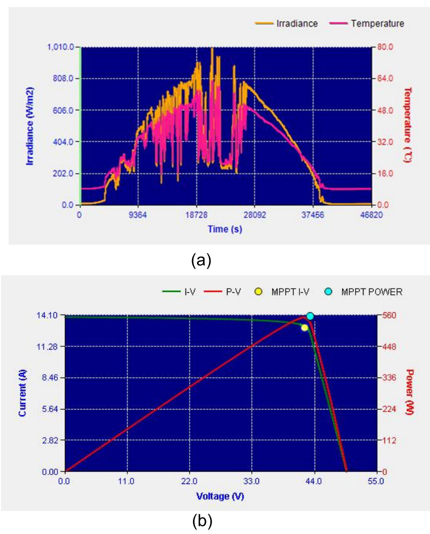

FIGURE 16. **(a)Irradiance and temperature variation curves under cloudy conditions(b) Host computer MPPT tracking curve.**

temperature and irradiance under these conditions, showing that both fluctuate significantly. Testing the MPPT algorithm with this curve helps validate its reliability. FIGURE 16(b) presents the MPPT tracking curve of the NSNPSO-INC algorithm under cloudy conditions. From the I-V and P-V curves in the figure, it is evident that the algorithm accurately tracks the MPP.

V. CONCLUSION AND OUTLOOK

This paper proposes a hybrid MPPT algorithm that effectively combines the population-based search advantages of the PSO algorithm with the rapid response capabilities of the INC algorithm. The particle velocity update process of the conventional PSO algorithm is simplified, and the iteration process is optimized using the principles of natural selection. The quick response characteristic of the INC method is then integrated to design the new algorithm. Simulation and experimental validation were conducted to compare the performance of different algorithms under various conditions, leading to the following conclusions:

1. Under static conditions without shading, all four algorithms accurately tracked the MPP. The tracking efficiency of the three different PSO algorithms stabilized around 99.9%, demonstrating superior performance compared to the conventional INC algorithm.

2. Under static partial shading conditions, the conventional INC algorithm fell into local MPP traps, whereas the three PSO algorithms accurately tracked the global MPP. The NSNPSO-INC algorithm converged within 0.14 seconds, improving convergence speed by 68.18% compared to the conventional PSO algorithm.

3. In dynamic irradiance changes under partial shading conditions, the NSNPSO-INC algorithm re-stabilized at the MPP within just 0.03 seconds, representing a 93.33% improvement in speed over the conventional PSO algorithm.

4. Experiments using a PV simulator to emulate PV array output showed that, compared to the conventional PSO algorithm, the NSNPSO-INC algorithm achieved the shortest convergence time, with a speed improvement of 85.52%. The power output remained stable around the maximum power of 550W, even under simulated cloudy conditions, accurately tracking the MPP.

Compared to the traditional PSO algorithm, the NSNPSO-INC algorithm offers significant enhancements by integrating the natural selection mechanism, which effectively improves the algorithm's global search capability. This allows the algorithm to escape local optima more efficiently and identify the global optimum with greater speed and accuracy. Furthermore, by combining the INC method, the NSNPSO-INC algorithm not only achieves faster convergence but also substantially improves the stability of power output. It is particularly robust in handling partial shading and rapid irradiance fluctuations, which are common challenges in PV systems.Additionally, the algorithm's low computational complexity and hardware requirements make it highly adaptable to a wide range of practical applications. It is especially suitable for use in remote areas, where reliable energy supplies are crucial, and in mobile devices, which benefit from its efficiency and adaptability. This work not only provides a powerful solution for PV MPPT but also introduces novel strategies that can be applied to other complex optimization problems in engineering domains.By addressing both stability and adaptability, the NSNPSO-INC algorithm presents a comprehensive approach that advances the state of MPPT algorithms and opens up new possibilities for optimizing systems in dynamic environments.

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